# Guarded summation-polynomial fall degrees

Study report, 28–29 October 2026. Derived bounds and development checks; no performance claim.

Let $S$ be $B$ distinct rational affine-point abscissas, $g(T) = \prod_{s\in S}(T - s)$, and $r = x(R)$ for an affine rational target. The ordinary polynomial-ring inputs are

$$g(x), \quad g(y), \quad g(z), \quad S_3(x, y, u), \quad S_3(u, z, r), \quad v(x - y) - 1.$$

Both point signs are allowed. The guard requires distinct first abscissas. Every degree below is ordinary total degree with explicit membership equations. The first-fall convention is homogeneous-top syzygies modulo Koszul syzygies.

| Setting, unsatisfiable instances | First fall | Last fall |
|---|---|---|
| Short Weierstrass, characteristic $> 3$, $B = 3$ | 4 | 5 |
| Same model, $B \geq 3$ | $B + 1$ | $\max(5, B + 1) \leq d_F \leq 3B - 4$ |
| Ordinary binary, $B = 3$, $g(T) = T^3 + c$ | 4 | 4 |
| Ordinary binary, $B = 3$, other split cubics | 4 | 5 |
| Ordinary binary, $B \geq 4$ | $B + 1$ | $B + 1 \leq d_F \leq 3B - 4$ |

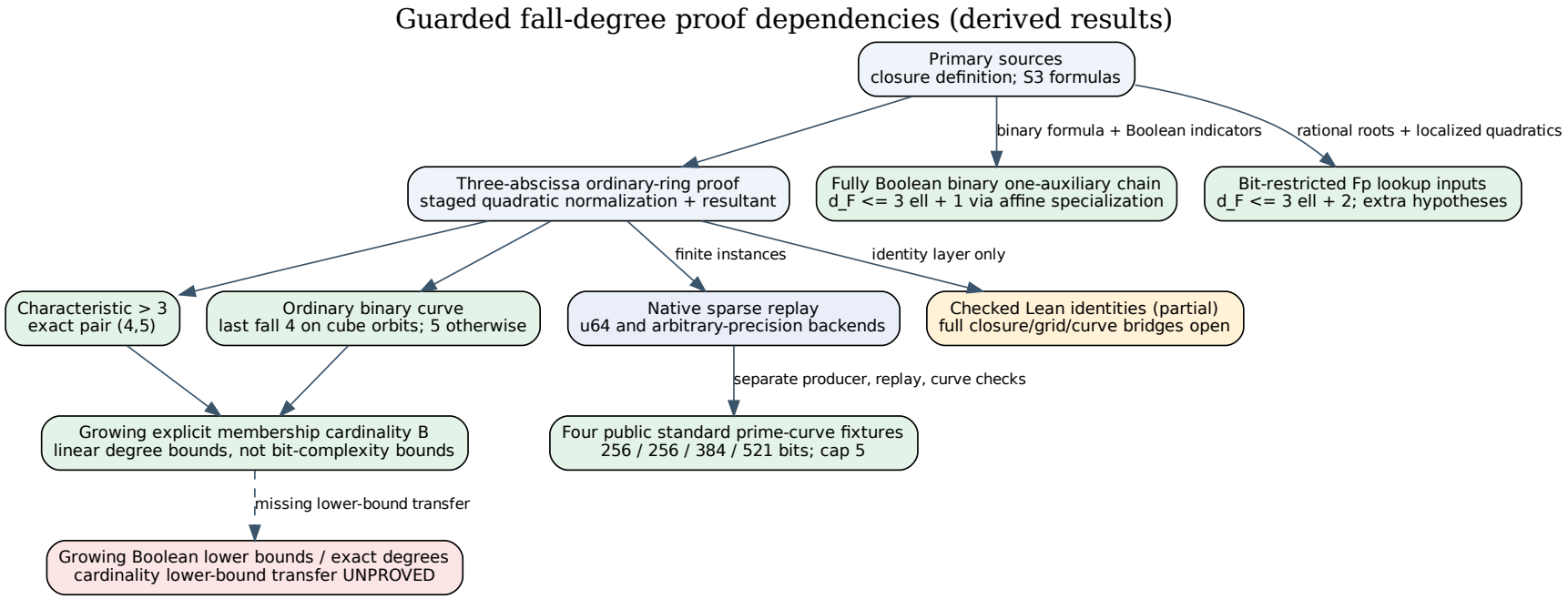
The binary curve is $Y^2 + XY = X^3 + a_2X^2 + a_6$, $a_6 \neq 0$. Its cube-orbit exception requires even extension degree, so the pair is $(4, 5)$ for all three-abscissa unsatisfiable instances over $\mathbb{F}_{2^n}$ with odd $n$. The growing lower bound depends on the degree-$B$ membership input and gives no lower bound in the variable count of a fixed-degree Boolean encoding.

**Direct Boolean upper bound.** For the binary two-$S_3$ chain with one shared $u$, three linear $\ell$-bit base embeddings, and every Boolean bit equation explicit, $d_F \leq 3\ell + 1$ for $\ell \geq 1$, independently of $n$. This holds with or without the inverse guard, without rational-lift assumptions. Fixing the base bits leaves an affine auxiliary system; staged assignment indicators localize its contradiction. Proof size may be exponential.

**Method.** Staged multiplication and cubic reduction normalize two quadratic fibres, then derive their scalar resultant within degree five. Finite-grid inversion yields 1. Characteristic two uses a Frobenius identity instead of division by two. Closed separating superspaces give the lower bounds. An exact triple-selector partition lifts local proofs to growing cardinality.

**Verification boundary.** Native sparse replay checks the compact historical proof DAGs against separately supplied inputs. Native binary-field arithmetic and the pinned cube-orbit fixture have development unit checks; no new field/base/target sweep or timing measurement was run. Lean checks only the declarations and hypotheses listed in `formal/PROOF_MAP.json`. The complete filtered-closure, grid-inversion, Boolean-localization, and elliptic-curve bridges are not formalized.

**Prior work.** The checked sources are Huang–Kosters–Yang–Yeo, 1505.02532; Caminata–Gorla, 2112.05579; Semaev, 1504.01175; and Kosters–Yeo, 1503.08001. Assumptions and repository links are in `SOURCES.md`. No literature-priority claim is made.




Solid arrows denote dependencies in the written arguments. The dashed arrow is an unresolved lower-bound transfer, not a proved implication. The Lean box certifies algebraic identity lemmas only; native replay certifies finite proof operations only. Neither substitutes for the complete mathematical derivation.

The existing E12 Boolean Weil-descent records are unchanged. Its completed cell retains labels $\geq 5, 4, \geq 5, 4$, a frozen lower median of 4, and a separate independent closure-replay gap. The present coefficient-field theorems do not close that gap. The direct Boolean upper theorem requires a separate encoding audit before application to that ledger. No performance or security claim follows.

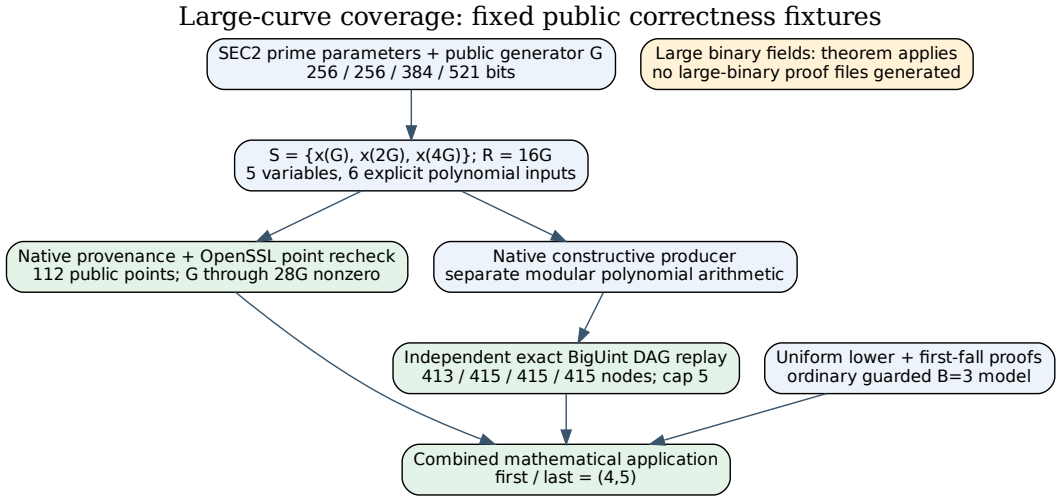
**Reproducible records.** See `README.md`, the proof files, `native/validation/`, `identities/`, and `formal/`. The editable diagram source is `reports/proof-dependencies.dot`. Historical Python sources are preserved as provenance, not as an execution path.

## Large standard-curve applications

Fixed public correctness fixtures; exact integer arithmetic, no timing comparison.

| Standard curve | Field bits | Replayed DAG nodes | Maximum degree | Point/input recheck |
|---|---|---|---|---|
| secp256k1 | 256 | 413 | 5 | accepted |
| secp256r1 / P-256 | 256 | 415 | 5 | accepted |
| secp384r1 / P-384 | 384 | 415 | 5 | accepted |
| secp521r1 / P-521 | 521 | 415 | 5 | accepted |

Every fixture uses $S = \{x(G), x(2G), x(4G)\}$ and $R = 16G$. Rechecking the nonzero public multiples $G, \ldots, 28G$ excludes both target signs by an integer scalar bound. OpenSSL 3.5.7 independently matches all 112 public points and the curve parameters. Producer and sparse replay use separate polynomial implementations; numeric JSON never passes through floating point.




Degree-five refutations are replayed computations. Exact pair $(4, 5)$ also uses the uniform lower/first-fall proofs, not a new degree-four closure. The five-variable, three-abscissa model is fixed; no full-DLP cost is measured. The large binary curves have theorem coverage only in this extension.

**Sources and records.** SEC2 v2.0, Sections 2.4–2.6; `extensions/large_curves/README.md`, the pinned parameter snapshot, and per-curve input/witness/DAG/replay records. Prime-field interpretation assumes the standard field primes. Full degree, curve-provenance and group-order arguments remain outside the Lean claim.

# Prime-field bit restrictions: a size-independent upper bound

Let $p > 3$, let the curve be nonsingular, and use three blocks of $\ell \geq 1$ Boolean-constrained base variables. Each coordinate lookup has degree at most $\ell$ and takes every Boolean assignment to a rational point abscissa. Encode $u$ and optional $v$ by independent $n$-bit integer maps onto $\mathbb{F}_p$, with $2^n \geq p$. The coefficient field remains $\mathbb{F}_p$. Include every bit equation $W^2 - W$ and supply the multilinear normal forms of the two summation equations and optional inverse guard.

$$\boxed{\text{UNSAT} \quad \Longrightarrow \quad d_F \leq 3\ell + 2.}$$

The cap is independent of the modulus size and of $n$. It depends on the exact generating presentation; it is not a theorem for a $\mathbb{F}_2$ carry-circuit implementation of prime-field arithmetic.

**Degree accounting.** Put $L = 3\ell$. Each normalized equation has base support $s_i \leq 2\ell$ and auxiliary degree at most two. Multiply by assignment-indicator factors on its support, reducing Boolean powers after each step. Every reduced row has degree at most $s_i + 2$; a next factor costs at most $s_i + 3 \leq L + 2$. The complementary factors then give the fully localized equation within $L + 2$.

After fixing the base assignment, rational point lifts make the first quadratic's roots rational. The equal-abscissa case is linear or a nonzero constant; an active inverse guard excludes it directly. Unsatisfiability therefore gives coprime univariate polynomials of degree at most two. Their constant refutation needs degree at most two: a monic difference gives $u - \alpha$, and subtracting its linear multiples from the quadratic leaves its nonzero value at $\alpha$. Carrying the assignment indicator throughout costs at most $L + 2$. Summing all indicators gives the exact identity 1.

**Raw inputs.** If the supplied substitutions have maximum degree $d_{\text{raw}}$, normalizing them first gives

$$d_F \leq \max(3\ell + 2, d_{\text{raw}}).$$

Arbitrary multilinear lookups have $d_{\text{raw}} \leq 4\ell$; affine lookups have $d_{\text{raw}} \leq 4$ and retain the $3\ell + 2$ bound.

**Limits.** Lookup tables and the proof may have exponentially many terms or rows. No first-fall equality, growing lower bound, exact degree, runtime gain, general ceiling five, or E12 certification follows. Eight local algebraic dependencies pass Lean build and fresh kernel replay, including the explicit discriminant/diagonal branch and quadratic cancellation. The complete filtered-closure theorem is not Lean formalized. The full statement and all edge cases are in `extensions/large_curves/prime_bit_restricted_upper.txt`.